

Sample Question Paper 1

Unit 1 — CPU Scheduling Algorithms

Q1(A) [5 Marks]

Consider the following processes with length of CPU burst time in milliseconds. All processes arrived at time zero in the order P1, P2, P3, P4.

Process	Burst Time
P1	5
P2	10
P3	2
P4	1

- Draw Gantt charts illustrating the execution of these processes for SJF and Round Robin (quantum = 1).
- Calculate waiting time for each process for each scheduling algorithm.
- Calculate average waiting time for each scheduling algorithm.

Q1(B) [3 Marks]

Differentiate between user mode and kernel mode in an OS. Why is it necessary?

Unit 2 — Process Synchronization & Deadlocks

Q2(A) [5 Marks]

Consider a system with 3 processes (P1, P2, P3) and 3 types of resources (R1, R2, R3). Available resources: (3, 3, 2). Allocated and Maximum Need matrices are given below:

Process	Allocation (R1, R2, R3)	Max Need (R1, R2, R3)
P1	(1, 0, 1)	(3, 2, 2)
P2	(2, 1, 1)	(4, 3, 3)
P3	(1, 1, 0)	(3, 2, 2)

- Calculate the Need Matrix.
- Determine if a request from P2 for (1, 0, 1) can be granted immediately without leading to an unsafe state.

Q2(B) [2 Marks]

Identify the characteristic requirements for the solutions to the critical section problem.

Unit 3 — Memory Management

Q3(A) [5 Marks]

Consider the following page reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1
Assume there are three frames. Apply the LRU page replacement algorithm to the reference string above and find out how many page faults are produced.

Q3(B) [3 Marks]

Summarize the differences between fixed and dynamic partitioning techniques.

Unit 4 — File Systems & Disk Scheduling

Q4(A) [5 Marks]

Consider a disk drive with 4,000 cylinders (numbered 0 to 3,999). Request queue: 1045, 750, 932, 878, 1365, 1787, 1245, 664, 1678, 1897. Current head position: 1167. Previous request served at: 1250.

Compute the total distance (in cylinders) that the disk arm would move for each of the following algorithms: FIFO, SSTF, SCAN, and C-SCAN.

Q4(B) [3 Marks]

What are the different directory structures used in operating systems? How do they impact file retrieval?

Sample Question Paper 2

Time: 1 Hour 30 Minutes | Maximum Marks: 32 | All questions are compulsory
Q(A) = 5 Marks (Numerical) | Q(B) = 2/3 Marks (Theory) | No Internal Choices

Unit 1 — CPU Scheduling Algorithms

Q1(A) [5 Marks]

Consider 3 processes P1, P2, and P3, which require 5, 7, and 4 time units and arrive at time 0, 1, and 3 respectively.

Draw the Gantt chart, process completion sequence, and average waiting time for:

- (i) Round Robin scheduling with CPU quantum of 2 time units.
- (ii) FCFS.

Q1(B) [3 Marks]

Examine the effects of preemptive and non-preemptive scheduling on process states.

Unit 2 — Process Synchronization & Deadlocks

Q2(A) [5 Marks]

Does the following pseudo code result in a deadlock? Justify your answer.

P1: lock(R1); lock(R2);

P2: lock(R2); lock(R3);

P3: lock(R3); lock(R1);

Analyze the synchronization solution, identify the possibility of deadlock, and design a deadlock-free version using any one prevention technique. Justify your approach.

Q2(B) [2 Marks]

Why does a deadlock occur in an operating system? Which are the four necessary conditions for a deadlock to occur?

Unit 3 — Memory Management

Q3(A) [5 Marks]

Calculate page faults for LRU, FIFO, and OPT for the following reference sequence with 3 page frames:

Reference string: 0, 1, 2, 1, 4, 2, 3, 7, 2, 1, 3, 5, 1, 2, 5

Show all steps and state the total page faults for each algorithm.

Q3(B) [3 Marks]

How logical addresses are translated into physical addresses in paging? Discuss the role of the MMU (Memory Management Unit).

Unit 4 — File Systems & Disk Scheduling

Q4(A) [5 Marks]

Given Request queue, initial head, disk size and direction. Compare performance of FCFS, SSTF, SCAN and C-SCAN.

Request queue: 176, 79, 34, 60, 92, 11, 41, 114, 7, 150, 130, 20

Initial head: 50 | Disk size: 0–199 | Direction: towards 199

Compute total cylinder movements for each algorithm.

Q4(B) [3 Marks]

Compare different file allocation methods: contiguous, linked, and indexed. What are their advantages and disadvantages?

Sample Question Paper 3

Unit 1 — CPU Scheduling Algorithms

Q1(A) [5 Marks]

Consider the following processes:

Process	Arrival Time	Burst Time
P1	0 ms	7 ms
P2	1 ms	5 ms
P3	2 ms	1 ms
P4	3 ms	2 ms

Compare Shortest Job First (SJF) — Preemptive and FCFS scheduling algorithms based on average waiting time. Draw the Gantt chart for each.

Q1(B) [2 Marks]

How does the Process Control Block (PCB) contribute to process switching in a multiprogramming system?

Unit 2 — Process Synchronization & Deadlocks

Q2(A) [5 Marks]

Race conditions are possible in many computer systems. Consider a banking system that maintains an account balance with two functions: deposit(amount) and withdraw(amount). These functions are passed the amount to be deposited or withdrawn.

Assume that a husband and wife share a bank account. Concurrently, the husband calls withdraw() and the wife calls deposit().

- Describe how a race condition is possible in this scenario.
- What might be done to prevent the race condition from occurring?

Q2(B) [3 Marks]

How does the Banker's Algorithm help in avoiding deadlocks? What are the limitations of using the Banker's Algorithm?

Unit 3 — Memory Management

Q3(A) [5 Marks]

A system has memory partitions of sizes 120 KB, 450 KB, 300 KB, 500 KB, and 700 KB. Four processes of sizes 350 KB, 275 KB, 150 KB, and 600 KB arrive for execution.

- Apply the Worst Fit memory allocation strategy.
- Calculate the total external fragmentation after allocation.

Q3(B) [2 Marks]

Discuss the role of the Translation Lookaside Buffer (TLB) in virtual memory management.

Unit 4 — File Systems & Disk Scheduling

Q4(A) [5 Marks]

Suppose that a disk drive has 5,000 cylinders, numbered 0 to 4,999. The drive is currently serving a request at cylinder 2,150 and the previous request was at cylinder 1,805.

Pending requests (FIFO order): 2069, 1212, 2296, 2800, 544, 1618, 356, 1523, 4965, 3681.

Calculate the total distance (in cylinders) the disk arm moves for: a) FCFS b) SCAN c) C-SCAN d) SSTF

Q4(B) [3 Marks]

How does free-space management work in file systems? Explain bitmap, linked-list, and grouping methods.

Sample Question Paper 4

Unit 1 — CPU Scheduling Algorithms

Q1(A) [5 Marks]

Consider the following processes with a time quantum of 3 ms:

Process	Arrival Time	Burst Time	Priority
P1	0 ms	5 ms	2
P2	1 ms	3 ms	1
P3	2 ms	8 ms	4
P4	3 ms	6 ms	3

Compare Non-Preemptive and Preemptive versions of Priority Scheduling based on average turnaround time. Draw the Gantt chart for each.

Q1(B) [3 Marks]

Compare multi-tasking, multi-programming and multi-processing Operating Systems.

Unit 2 — Process Synchronization & Deadlocks

Q2(A) [5 Marks]

Consider the following two threads T1 and T2 that update two shared variables a and b. Assume that initially $a = b = 1$. Though context switching can happen at any time, each statement is executed atomically without interruption.

T1: $a = a + 1$; $b = b + 1$;

T2: $b = 2 * b$; $a = 2 * a$;

List ALL possible combinations of values of a and b after both T1 and T2 finish execution.

Q2(B) [2 Marks]

Compare semaphore and mutex. When should each be used?

Unit 3 — Memory Management

Q3(A) [5 Marks]

A system has a TLB with an access time of 25 ns. The main memory access time is 150 ns. If the TLB hit ratio is 85%, calculate the Effective Access Time (EAT). Show all steps.

Additionally: A system has a 32-bit virtual address space with a page size of 8 KB. Physical memory size is 512 MB.

a) Calculate the number of pages in the virtual address space.

b) Determine the number of bits required for the page number and the offset.

Q3(B) [3 Marks]

Compare paging and segmentation with respect to memory management. Discuss advantages and disadvantages of each.

Unit 4 — File Systems & Disk Scheduling

Q4(A) [5 Marks]

Given Request queue, initial head, disk size and direction. Compare performance of FCFS, SSTF, SCAN and C-SCAN.

Request queue: 10, 180, 20, 170, 30, 160, 40, 150, 50, 140, 60, 130

Initial head: 75 | Disk size: 0–199 | Direction: towards 0

Compute total cylinder movements and identify the most efficient algorithm.

Q4(B) [3 Marks]

How does the SCAN (Elevator Algorithm) scheduling method work? Why is it preferred over SSTF?

Sample Question Paper 5

Unit 1 — CPU Scheduling Algorithms

Q1(A) [5 Marks]

Consider a single processor system with four processes A, B, C, and D. The first value is arrival time and the second is CPU burst time:

A (0, 10), B (2, 6), C (4, 3), D (6, 7)

Compare Shortest Remaining Time First (SRTF) — Preemptive and Non-Preemptive algorithms as CPU scheduling policy. Draw Gantt charts and calculate average waiting time. Ignore context switching times.

Q1(B) [2 Marks]

Describe time-sharing Operating Systems with a real-life example.

Unit 2 — Process Synchronization & Deadlocks

Q2(A) [5 Marks]

Consider a multi-threaded program with two threads T1 and T2. The threads share two semaphores: s1 (initialized to 1) and s2 (initialized to 0). They also share a global variable x (initialized to 0).

Thread T1: wait(s1); x = x + 1; print(x); signal(s2); signal(s1);

Thread T2: wait(s1); x = x + 1; print(x); wait(s2); signal(s1);

Present which outcomes are possible when threads T1 and T2 execute concurrently. Justify each possible outcome.

Q2(B) [3 Marks]

How does an operating system detect a deadlock? How often should deadlock detection be performed in a system?

Unit 3 — Memory Management

Q3(A) [5 Marks]

Free memory holes of sizes 15K, 10K, 5K, 25K, 30K, 40K are available. Processes of size 12K, 2K, 25K, 20K are to be allocated.

Show how processes are placed using First Fit, Best Fit, and Worst Fit strategies.

Calculate internal as well as external fragmentation for each strategy.

Q3(B) [2 Marks]

Illustrate the concept of swapping and explain its role in memory management.

Unit 4 — File Systems & Disk Scheduling

Q4(A) [5 Marks]

A disk of size 512 MB is divided into blocks of 64 KB. A file is stored using linked allocation. In linked allocation, each data block reserves 4 bytes to store the pointer to the next data block. The link part of the last data block contains a NULL pointer (also of 4 bytes).

A file of 1 MB needs to be stored on the disk. Assume $1K = 2^{10}$ and $1M = 2^{20}$.

Calculate the amount of space (in bytes) that will be wasted due to internal fragmentation.

Q4(B) [3 Marks]

How does C-SCAN (Circular SCAN) differ from SCAN? What is its advantage in terms of disk head movement fairness?